organic compound using Fischer projection formula. Fischer has devised a projection formula to relate the structure of a carbohydrate to one of the two enantiomeric forms of glyceraldehyde (Figure 14.2). Based on these structures, carbohydrates are named as D or L.The carbohydrates are usually named with two prefixes namely D or L and followed by sign either (+) or (-). Carbohydrates are assigned the notation (D/L) by comparing the configuration of the carbon that is attached to -CH_2OH group with that of glyceraldehyde.For example D-glucose is so named because the H and OH on C5 carbon are in the same configuration as the H and OH on C2 carbon in D-Glyceraldehyde.

There + and – sign indicates the dextro rotatory and levo rotatory respectively. Dextro rotatory compounds rotate the plane of plane polarised light in clockwise direction while the levo rotatory compounds rotate in anticlockwise direction. The D or L isomers can either be dextro or levo rotatory compounds. Dextro rotatory compounds are represented as D-(+) or L-(+) and the levo rotatory compounds as D-(–) or L-(–)

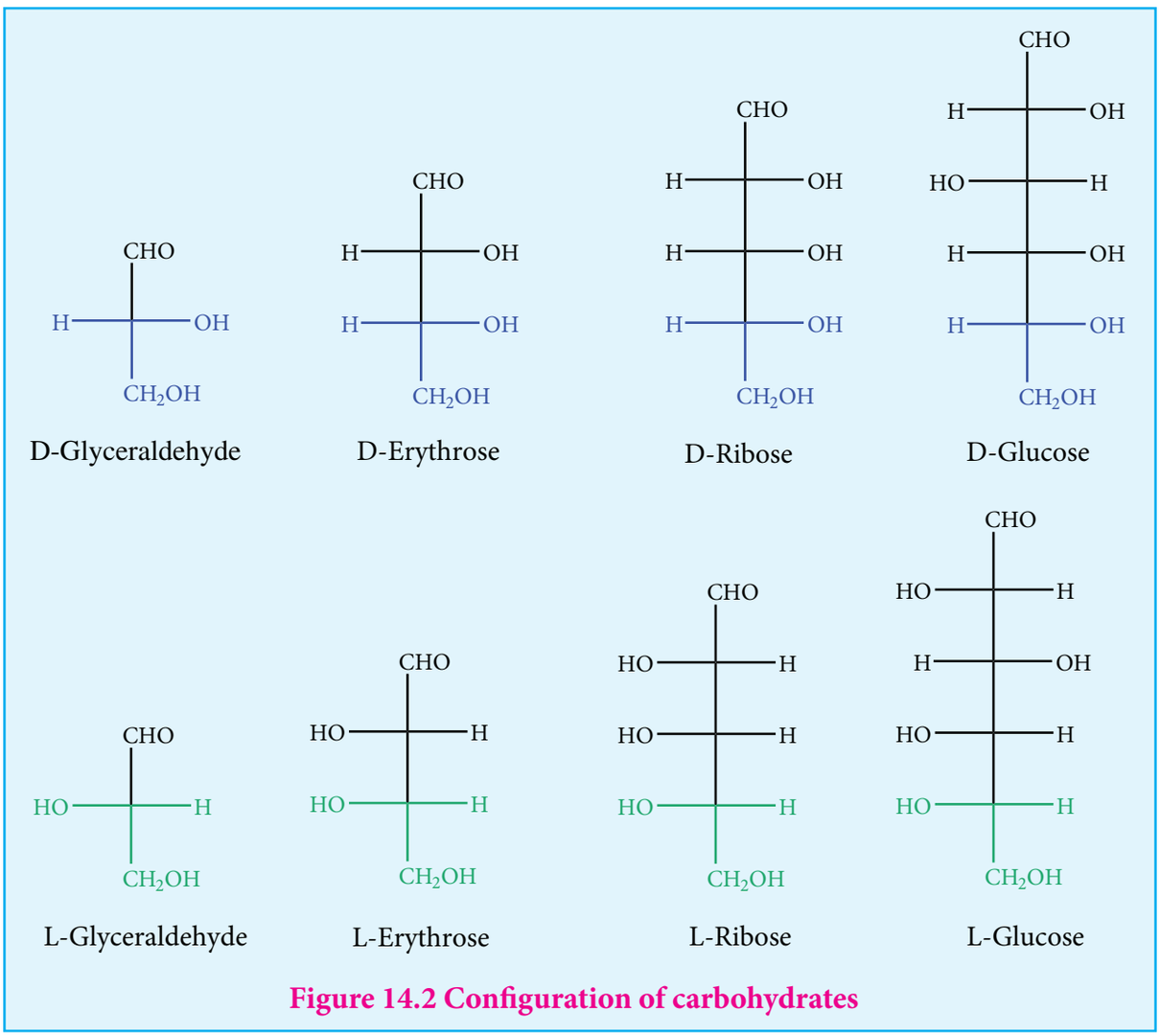


Figure 14.2 Configuration of carbohydrates

14.1.2 Classification of carbohydrates:

Carbohydrates can be classified into three major groups based on their product of hydrolysis, namely monosaccharides, oligosaccharides and polysaccharides.

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